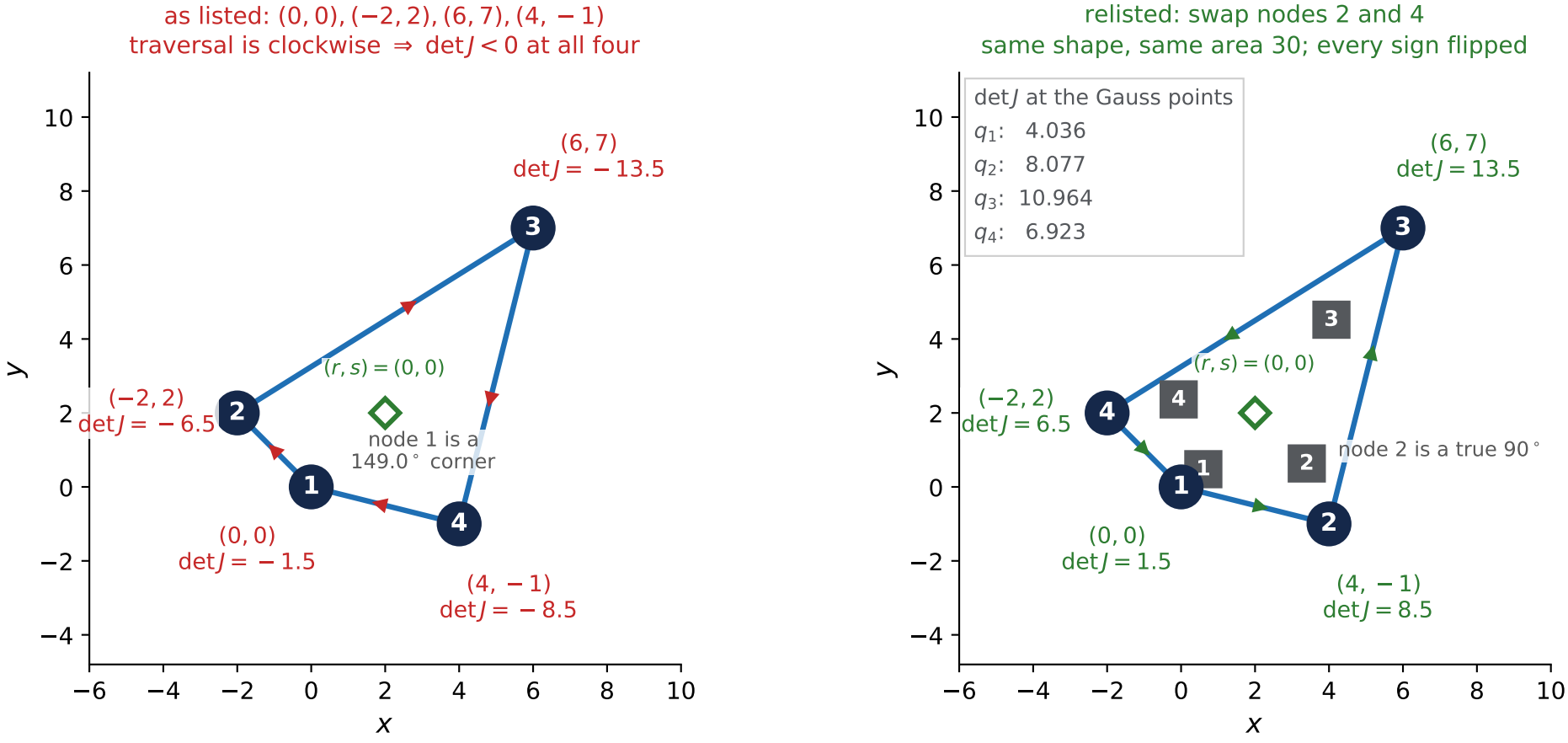

The four points are a perfectly good convex quadrilateral --- the element is not broken, the node list is.
All four det$J$ carry the *same* sign, which is the signature of a reversed list, not of a tangled element (mixed signs would be that).

Relisted, $\det J = \frac{15}{2} + \frac{7r}{2} + \frac{5s}{2}$: corners $\frac{3}{2}$, $\frac{17}{2}$, $\frac{27}{2}$, $\frac{13}{2}$, so max/min = 9 --- valid, but still a distorted element.